

Chapter 3

Data Preprocessing

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Outline

- Why preprocess the data?
- Major tasks in data preprocessing
 - Data cleaning
 - Data Integration
 - Data Reduction
 - Data Transformation

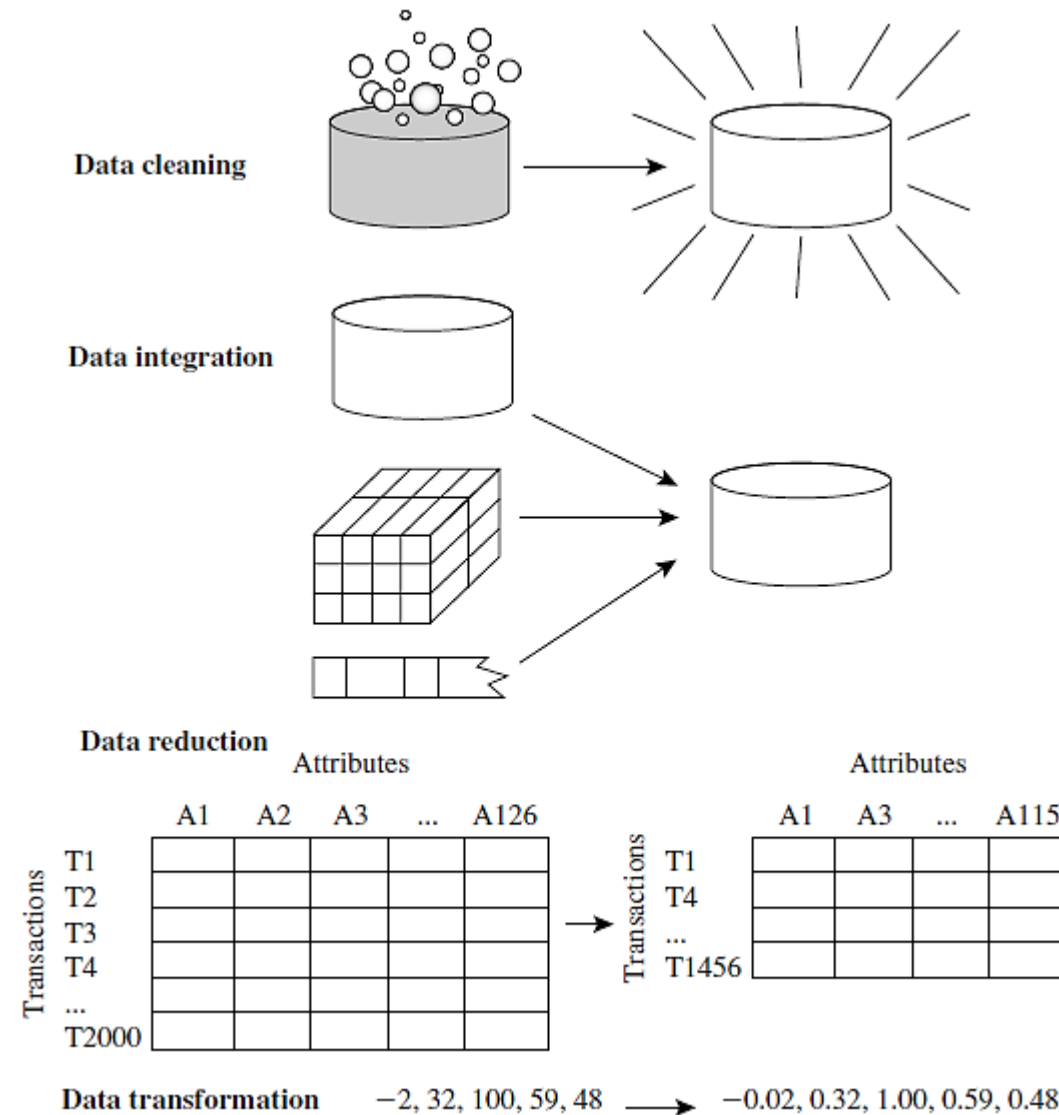
Why Data Preprocessing?

- Data in the real world is dirty
 - **incomplete**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., occupation= " "
 - **noisy**: containing errors or outliers
 - e.g., Salary="-10"
 - **inconsistent**: containing discrepancies in codes or names
 - e.g., Age="42" Birthday="03/07/1997"
 - e.g., Was rating "1,2,3", now rating "A, B, C"
- No quality data, no quality mining results!

Major Tasks in Data Preprocessing

- Data cleaning
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
 - Integration of multiple databases, data cubes, or files
- Data transformation
 - Normalization and aggregation
- Data reduction
 - Obtains reduced representation in volume but produces the same or similar analytical results

Forms of Data Preprocessing



Data Cleaning

- Data cleaning tasks
 - Fill in missing values
 - Identify outliers
 - Smooth out noisy data
 - Correct inconsistent data
 - Resolve redundancy caused by data integration

Missing Data

- Missing data may be due to
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (assuming the tasks in classification—not effective when the percentage of missing values per attribute varies considerably).
- Fill in the missing value manually: tedious + infeasible?
- Fill in it automatically with
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean
 - the attribute mean for all samples belonging to the same class: smarter
 - the most probable value: inference-based such as Bayesian formula or decision tree

Data Cleaning

- Find and correct misspelled words
- Detecting entry errors on simple variables with a frequency table

Value	Count
Good	1598647
Bad	1354468
Godo	15
Bade	1

correct

```
if x == "Godo":  
    x = "Good"  
if x == "Bade":  
    x = "Bad"
```

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may due to
 - faulty data collection instruments
 - data entry problems
 - data transmission problems

How to Handle Noisy Data?

- Binning
 - first sort data and partition into (equal-frequency) bins
 - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Regression
 - smooth by fitting the data into regression functions
- Clustering
 - detect and remove outliers
- Combined computer and human inspection
 - detect suspicious values and check by human (e.g., deal with possible outliers)

Data Smoothing Method: Binning

- **Equal-depth** (frequency) partitioning
 - Divides the range into N intervals, each containing approximately same number of samples
- **Equal-width** (distance) partitioning
 - Divides the range into N intervals of equal size: uniform grid
 - if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
 - $\text{Bin1}=[\text{min}, \text{min}+w)$, $\text{Bin2}=[\text{min}+w, \text{min}+2w)$, ... $\text{BinN}=[\text{min}+(N-1)w, \text{max}]$
 - The most straightforward, but outliers may dominate presentation
 - Skewed data is not handled well

Data Smoothing Method: Binning

❑ Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

❑ Partition them into 3 bins by **equal-frequency**

* Partition into bins:

- Bin 1: 4, 8, 9, 15
- Bin 2: 21, 21, 24, 25
- Bin 3: 26, 28, 29, 34

* Smoothing by bin means:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

* Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

Data Smoothing Method: Binning

❑ Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

❑ Partition them into 3 bins by **equal-width**

$$W = (34 - 4) / 3 = 10$$

- Bin1=[4,14), Bin2=[14, 24), Bin3=[24,34]

* Partition into bins:

- Bin 1: 4, 8, 9
- Bin 2: 15, 21, 21
- Bin 3: 24, 25, 26, 28, 29, 34

* Smoothing by bin means:

- Bin 1: 7, 7, 7
- Bin 2: 19, 19, 19
- Bin 3: 28, 28, 28, 28, 28, 28

Regression

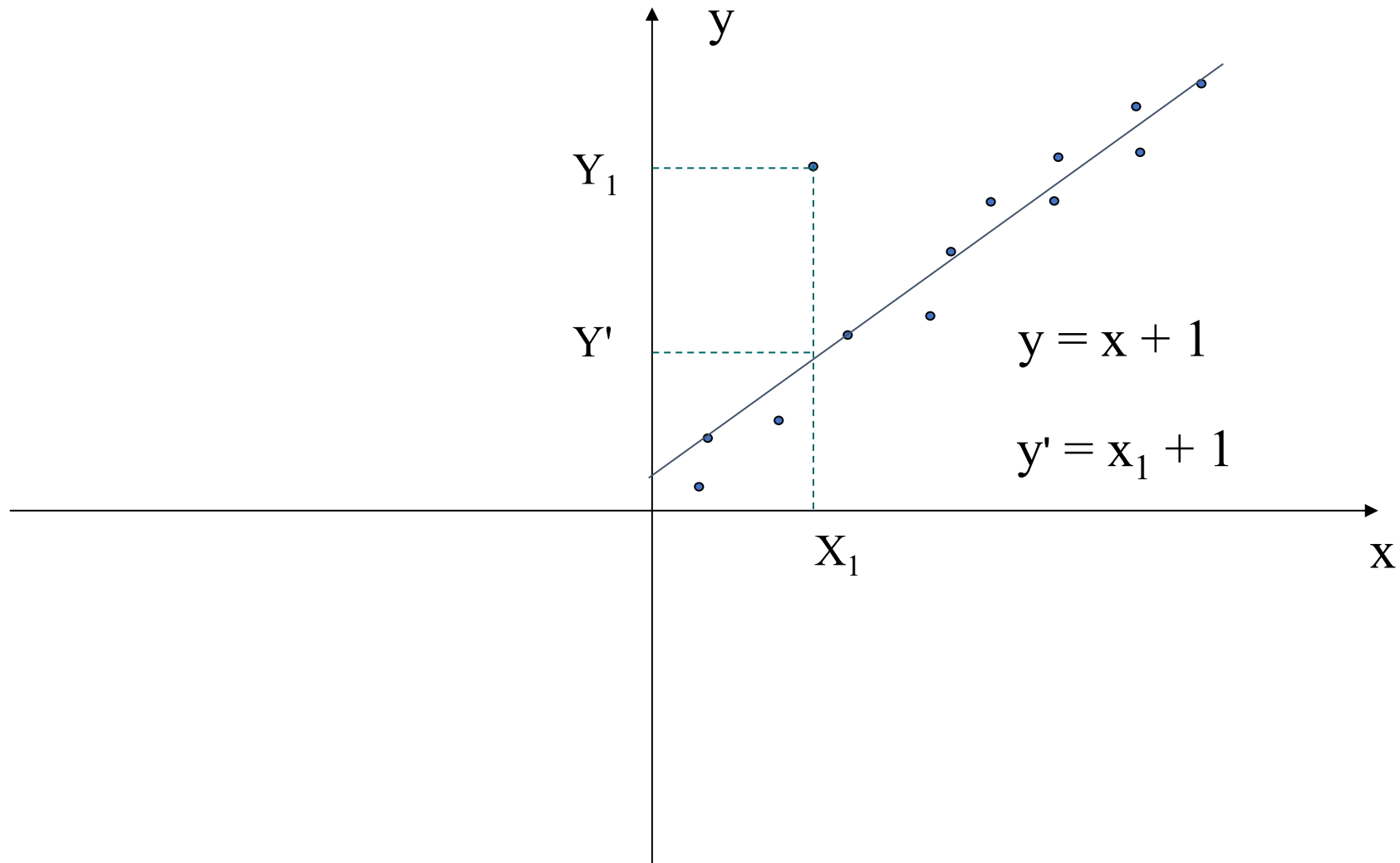
- Linear Regression

$$Y = \alpha + \beta X$$

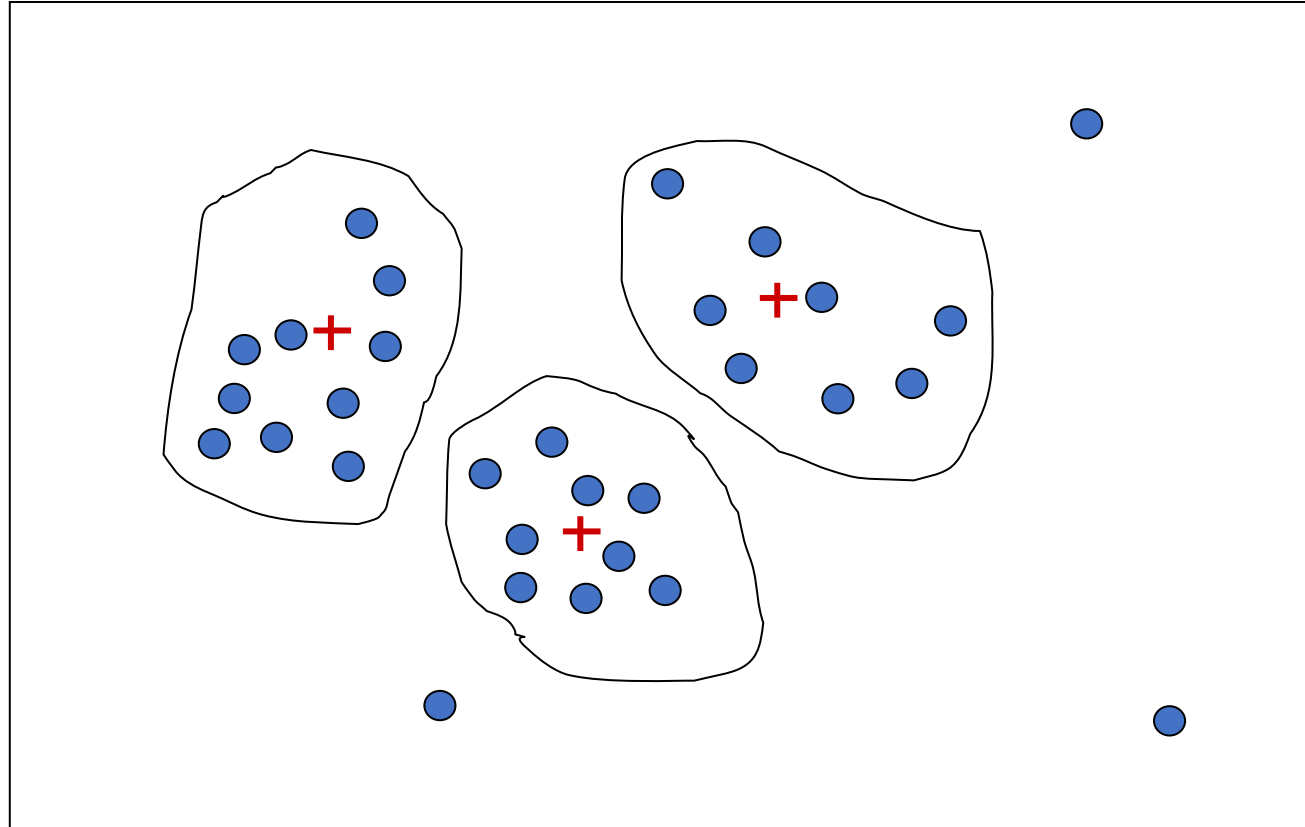
- Multiple Linear Regression

$$Y = b_0 + b_1 X_1 + b_2 X_2 + \dots + b_m X_m$$

Regression



Cluster Analysis

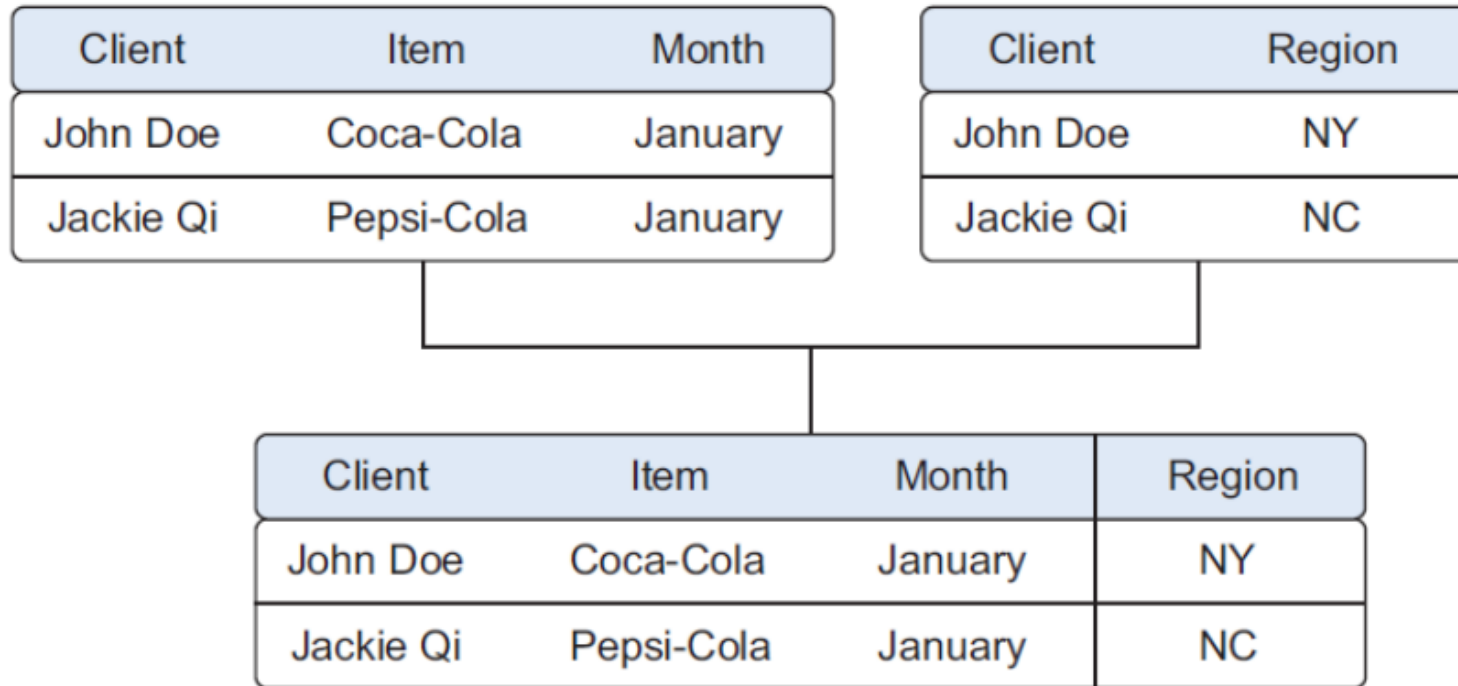


Data Integration

- Data integration:
 - Combines data from multiple sources into a coherent store
- Two operations to combine information from different data set
 - Joining tables
 - Appending tables

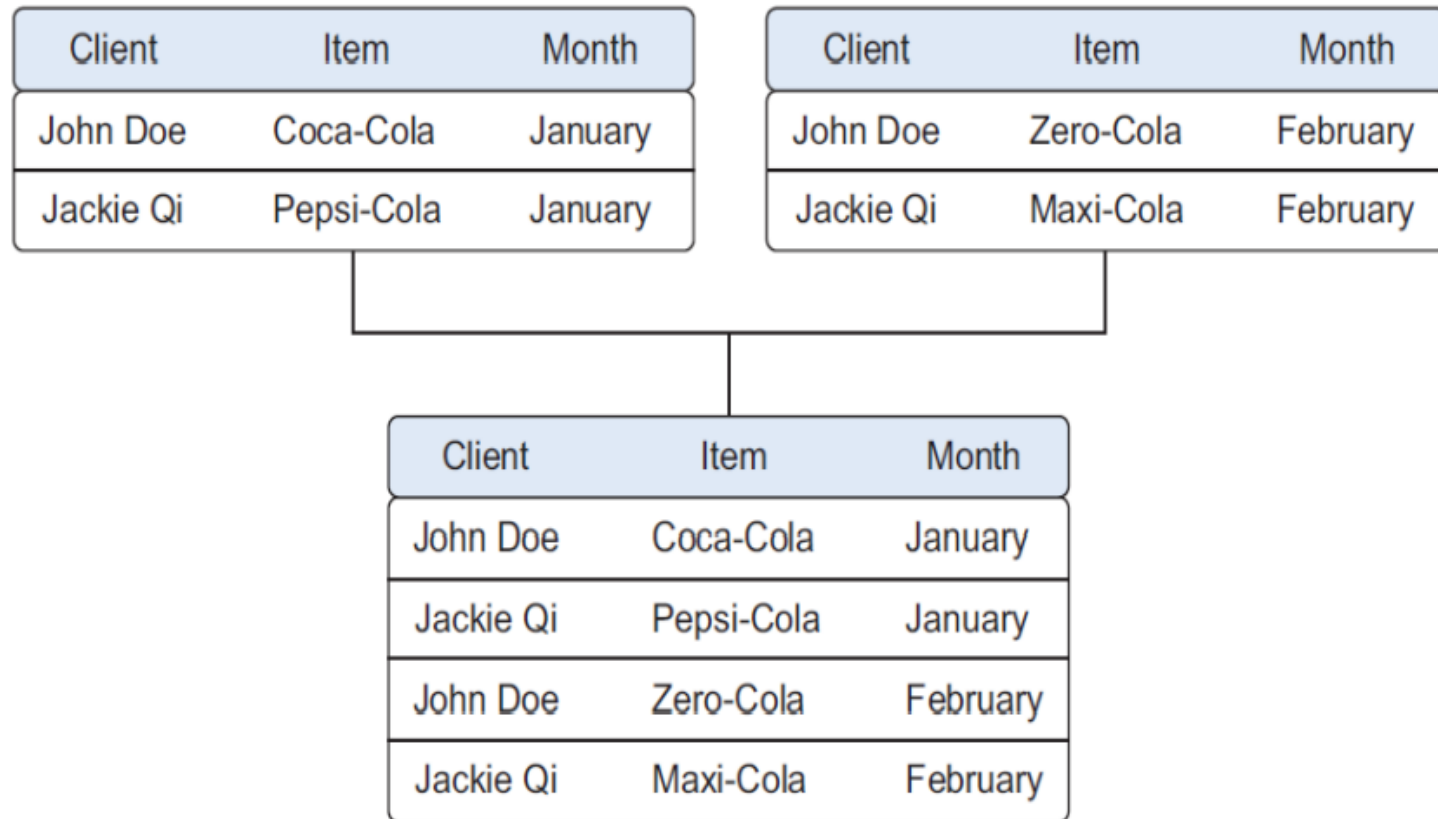
Data Integration

- Joining tables



Data Integration

- Appending tables



Data Integration

- Data integration may cause
 - Data Redundancy
 - Data Inconsistency
- Cause of data inconsistency
 - Encoding standard
 - attribute values from different sources are different
 - Unit measurement
 - different scales, e.g., metric vs. British units
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
 - *Object identification*: The same attribute or object may have different names in different databases
 - *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- Redundant attributes may be able to be detected by *correlation analysis*

Correlation Analysis (Numerical Data)

- Correlation coefficient (also called Pearson's product moment coefficient)

$$r_{A,B} = \frac{\sum(A - \bar{A})(B - \bar{B})}{(n - 1)\sigma_A\sigma_B}$$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective means of A and B , σ_A and σ_B are the respective standard deviation of A and B

- If $r_{A,B} > 0$, A and B are positively correlated (A 's values increase as B 's). The higher, the stronger correlation.
- $r_{A,B} = 0$: independent; $r_{A,B} < 0$: negatively correlated

Correlation Analysis (Categorical Data)

- χ^2 (chi-square) test

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where c is the number of distinct values of attribute A

r is the number of distinct values of attribute B

o_{ij} is the observed frequency

e_{ij} is the expected frequency of (A_i, B_j)

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{N}$$

- The larger the χ^2 value, the more likely the variables are related
- The χ^2 statistic tests the hypothesis that A and B are independent
- Degree of freedom (df) = $(r-1) \times (c-1)$

Chi-Square Calculation: An Example

	<i>male</i>	<i>female</i>	<i>Total</i>
<i>fiction</i>	250 (90)	200 (360)	450
<i>non_fiction</i>	50 (210)	1000 (840)	1050
Total	300	1200	1500

$$e_{11} = \frac{300 \times 450}{1500} = 90$$

- χ^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

Chi-Square Calculation: An Example

Probability level (alpha)						
df	0.5	0.10	0.05	0.02	0.01	0.001
1	0.455	2.706	3.841	5.412	6.635	10.827
2	1.386	4.605	5.991	7.824	9.210	13.815
3	2.366	6.251	7.815	9.837	11.345	16.268
4	3.357	7.779	9.488	11.668	13.277	18.465
5	4.351	9.236	11.070	13.388	15.086	20.517

$$df = (r-1) \times (c-1) = (2-1) \times (2-1) = 1$$

Significance level = 0.001

$$t = 10.827 \quad \chi^2 = 507.93$$

$\chi^2 > t$ then reject the hypothesis

- It shows that *gender* and *preferred_reading* are (strongly) correlated for the given group of people

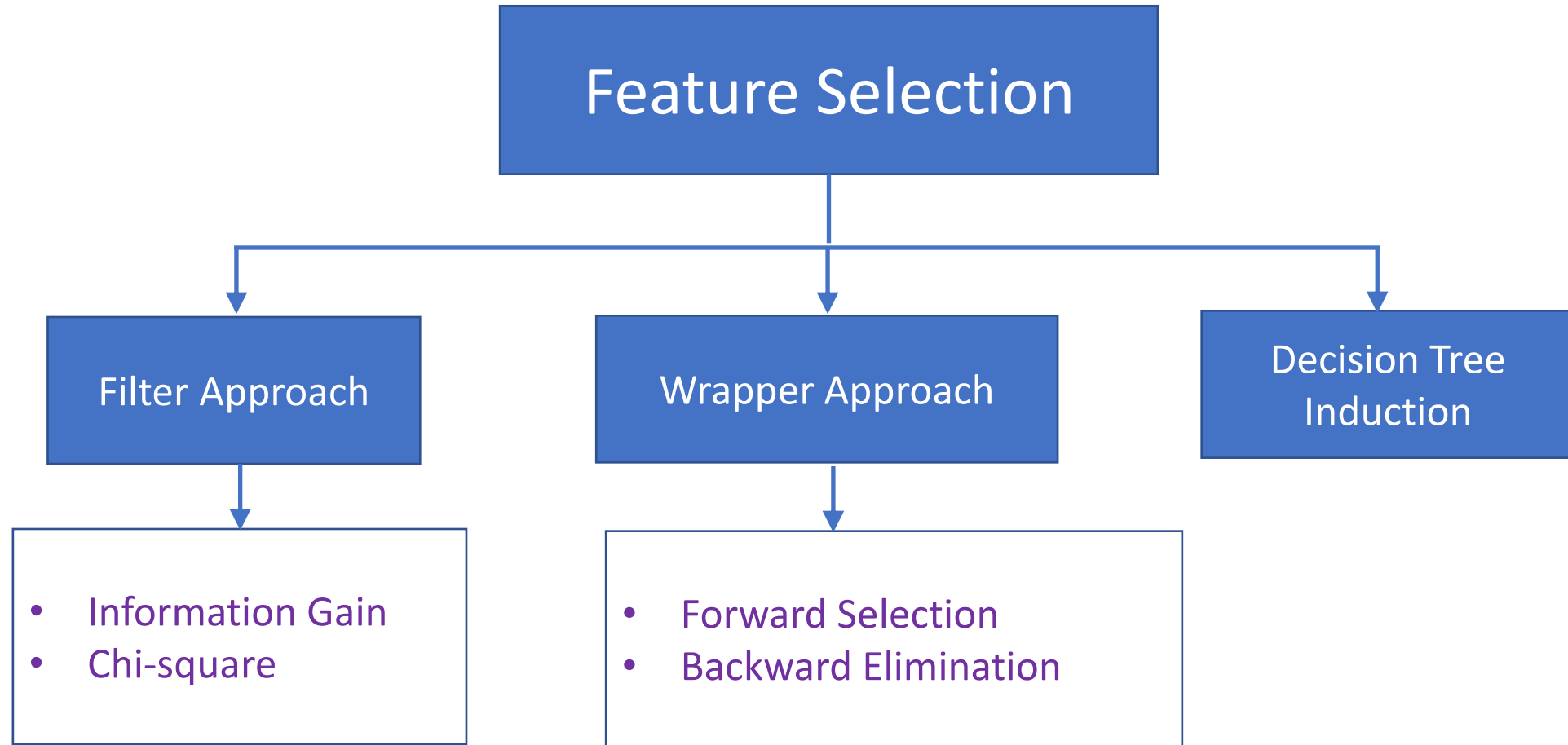
Data Reduction

- Why data reduction?
 - A database/data warehouse may store terabytes of data
 - Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction
 - Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- Data reduction strategies
 - Dimensionality reduction or feature selection
 - Numerosity reduction

Dimensionality Reduction

- Attribute Subset Selection
- Irrelevant, weakly relevant, or redundant attributes may be detected and removed
- Heuristic methods
 - Filter Approach
 - Wrapper Approach
 - Decision-tree induction

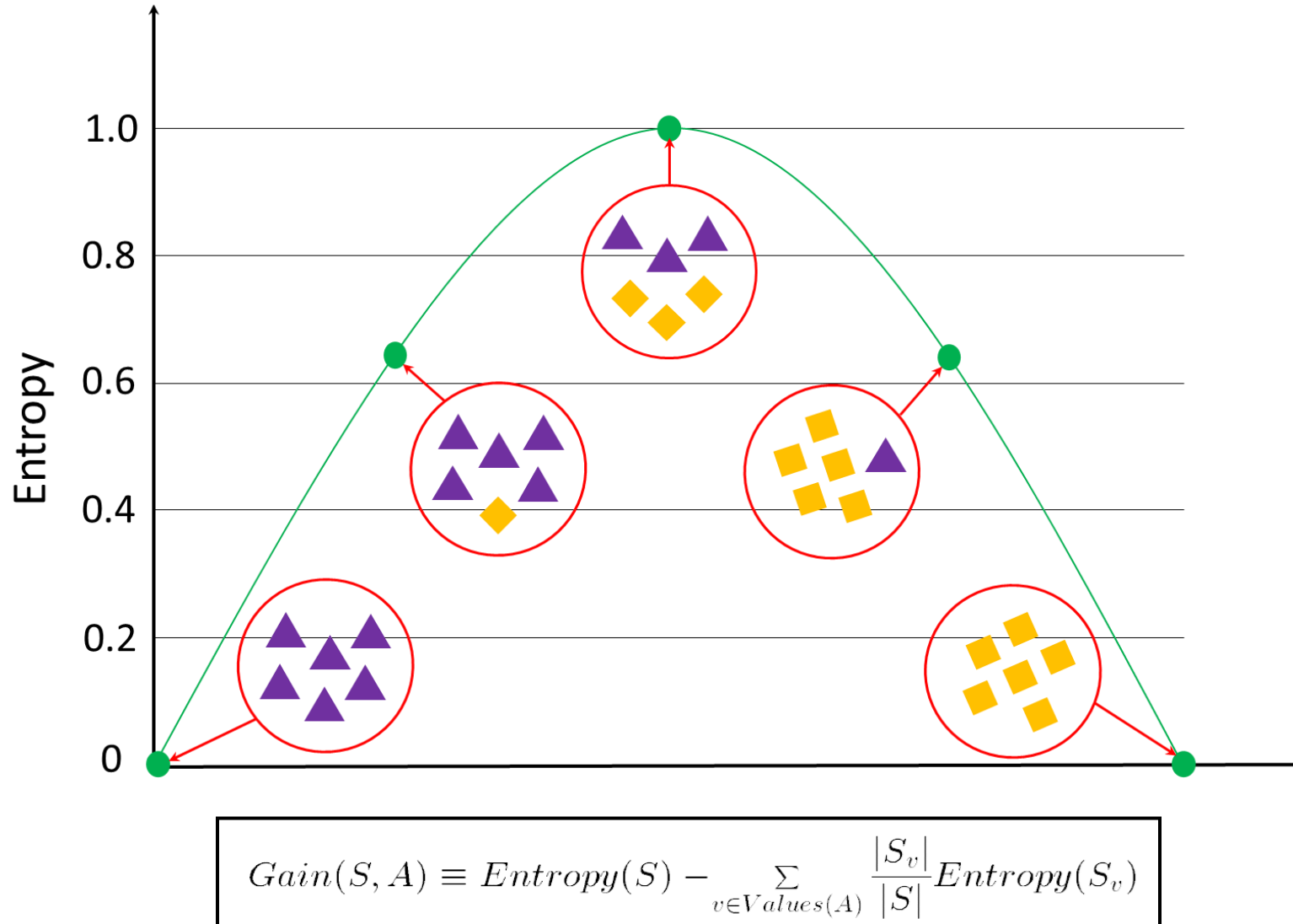
Feature Selection



Feature Selection

- Filter approach
 - Calculate weight of each attribute
 - Select the important attributes (usually the higher weight the more important attribute)
 - Example methods
 - Information Gain
 - Chi-square

Information Gain: IG



Information Gain: IG

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



attribute	IG
Outlook	0.247
Temperature	
Humidity	
Windy	

ตารางค่า Information Gain

Information Gain: IG

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



attribute	IG
Outlook	0.247
Temperature	0.029
Humidity	
Windy	

ตารางค่า Information Gain

Information Gain: IG

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



attribute	IG
Outlook	0.247
Temperature	0.029
Humidity	0.152
Windy	

ตารางค่า Information Gain

Information Gain: IG

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



attribute	IG
Outlook	0.247
Temperature	0.029
Humidity	0.152
Windy	0.048

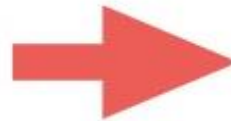
ตารางค่า Information Gain

Information Gain: IG

- Selected attributes: $IG > 0.1$

attribute	IG
Outlook	0.247
Humidity	0.152
Windy	0.048
Temperature	0.029

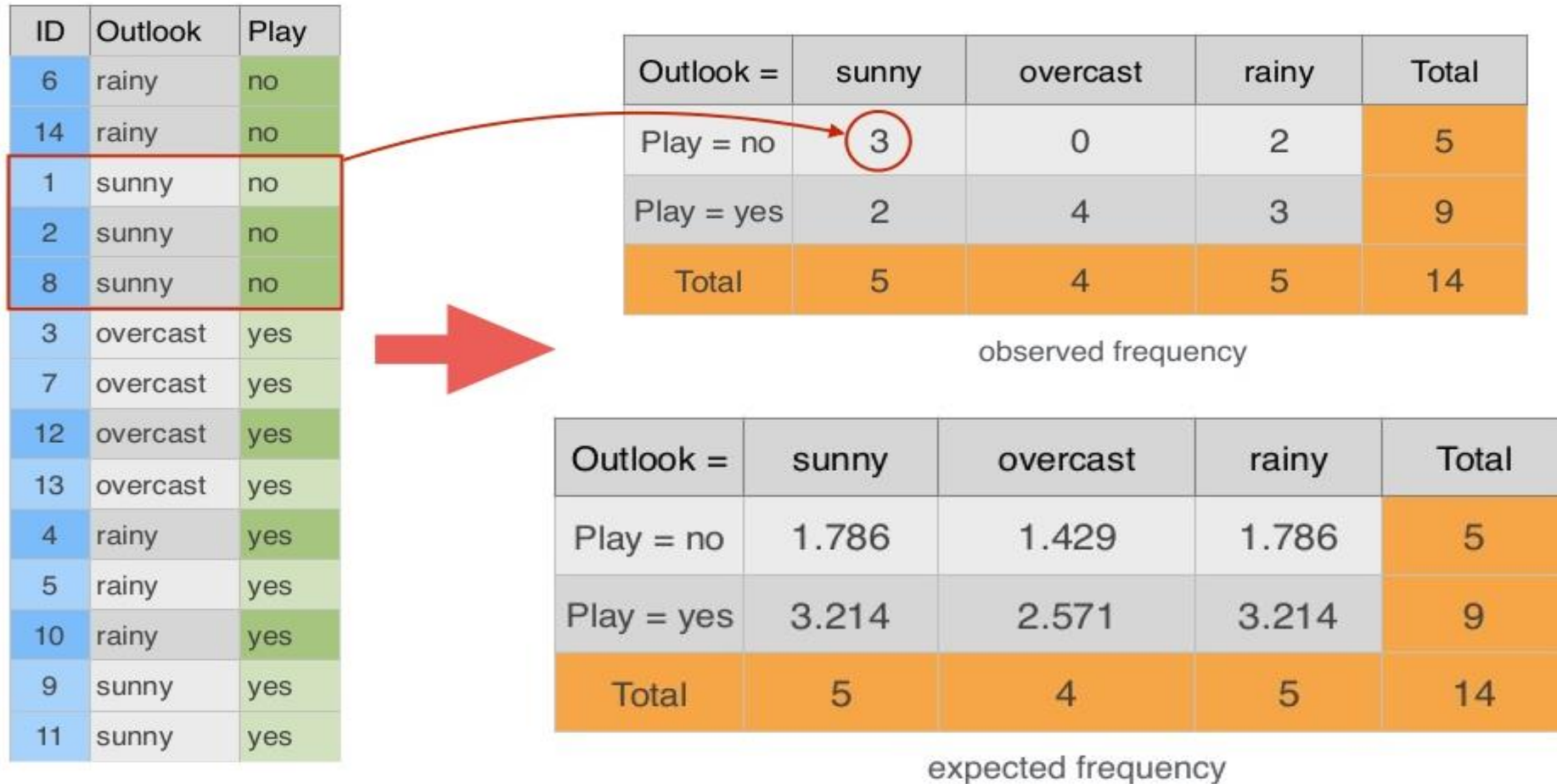
ตารางค่า Information Gain



ID	Outlook	Humidity	Play
1	sunny	high	no
2	sunny	high	no
3	overcast	high	yes
4	rainy	high	yes
5	rainy	normal	yes
6	rainy	normal	no
7	overcast	normal	yes
8	sunny	high	no
9	sunny	normal	yes
10	rainy	normal	yes
11	sunny	normal	yes
12	overcast	high	yes
13	overcast	normal	yes
14	rainy	high	no

Chi-square

- Compute correlation between Outlook and Play attributes



ID	Outlook	Play
6	rainy	no
14	rainy	no
1	sunny	no
2	sunny	no
8	sunny	no
3	overcast	yes
7	overcast	yes
12	overcast	yes
13	overcast	yes
4	rainy	yes
5	rainy	yes
10	rainy	yes
9	sunny	yes
11	sunny	yes

Outlook =	sunny	overcast	rainy	Total
Play = no	3	0	2	5
Play = yes	2	4	3	9
Total	5	4	5	14

observed frequency

Outlook =	sunny	overcast	rainy	Total
Play = no	1.786	1.429	1.786	5
Play = yes	3.214	2.571	3.214	9
Total	5	4	5	14

expected frequency

Chi-square

- Compute correlation between Outlook and Play attributes

Outlook =	sunny	overcast	rainy
Play = no	3	0	2
Play = yes	2	4	3

observed frequency

Outlook =	sunny	overcast	rainy
Play = no	1.786	1.429	1.786
Play = yes	3.214	2.571	3.214

expected frequency

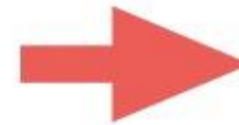
$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

- Chi-Square = $(3-1.786)^2/1.786 +$
 $(0-1.429)^2/1.429 +$
 $(2-1.786)^2/1.786 +$
 $(2-3.214)^2/3.214 +$
 $(4-2.571)^2/2.571 +$
 $(3-3.214)^2/3.214$

 $= 3.547$

Chi-square

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

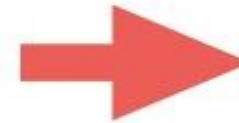


attribute	Chi-Square
Outlook	3.547
Temperature	
Humidity	
Windy	

ตารางค่า Chi-Square

Chi-square

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

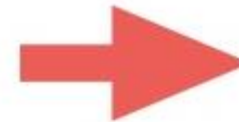


attribute	Chi-Square
Outlook	3.547
Temperature	0.570
Humidity	
Windy	

ตารางค่า Chi-Square

Chi-square

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

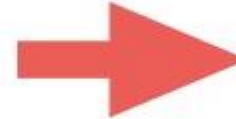


attribute	Chi-Square
Outlook	3.547
Temperature	0.570 0.94
Humidity	2.800
Windy	

ตารางค่า Chi-Square

Chi-square

ID	Outlook	Temperature	Humidity	Windy	Play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	mild	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	mild	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no



attribute	Chi-Square
Outlook	3.547
Temperature	0.570 0.94
Humidity	2.800
Windy	0.933

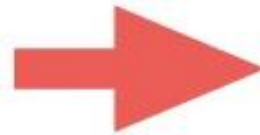
ตารางค่า Chi-Square

Chi-square

attribute	Chi-Square
Outlook	3.547
Humidity	2.800
Windy	0.933
Temperature	0.570 0.94

ตารางค่า Chi-Square

Selected attributes: Chi-square > 2.0



ID	Outlook	Humidity	Play
1	sunny	high	no
2	sunny	high	no
3	overcast	high	yes
4	rainy	high	yes
5	rainy	normal	yes
6	rainy	normal	no
7	overcast	normal	yes
8	sunny	high	no
9	sunny	normal	yes
10	rainy	normal	yes
11	sunny	normal	yes
12	overcast	high	yes
13	overcast	normal	yes
14	rainy	high	no

Feature Selection

- Wrapper approach
 - Calculate the weight of the classification model
 - Example methods
 - Forward Selection
 - Backward Elimination

Forward Selection

- Starts with an empty set of attributes as the reduced set
- The best of the original attributes is determined and added to the reduced set
- At each subsequent iteration or step, the best of the remaining original attributes is added to the set

Forward Selection

ID	Free	Won	Cash	Type
1	Y	Y	Y	spam
2	N	Y	Y	spam
3	N	N	N	normal
4	N	N	N	normal
5	Y	N	N	spam
6	Y	N	N	spam
7	N	N	N	normal
8	N	Y	N	spam
9	N	N	N	normal
10	N	N	N	normal



ID	Free	Type
1	Y	spam
2	N	spam
3	N	normal
4	N	normal
5	Y	spam
6	Y	spam
7	N	normal
8	N	spam
9	N	normal
10	N	normal

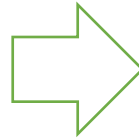


Accuracy =
80%

Attribute	Accuracy
Free	80%
Won	
Cash	

Forward Selection

ID	Won	Type
1	Y	spam
2	Y	spam
3	N	normal
4	N	normal
5	N	spam
6	N	spam
7	N	normal
8	Y	spam
9	N	normal
10	N	normal

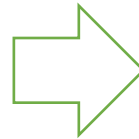


Accuracy = 80%

Attribute	Accuracy
Free	80%
Won	80%
Cash	

Forward Selection

ID	Cash	Type
1	Y	spam
2	Y	spam
3	N	normal
4	N	normal
5	N	spam
6	N	spam
7	N	normal
8	N	spam
9	N	normal
10	N	normal

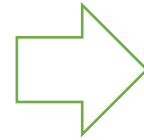


Accuracy = 70%

Attribute	Accuracy
Free	80%
Won	80%
Cash	70%

Forward Selection

ID	Free	Won	Type
1	Y	Y	spam
2	N	Y	spam
3	N	N	normal
4	N	N	normal
5	Y	N	spam
6	Y	N	spam
7	N	N	normal
8	N	Y	spam
9	N	N	normal
10	N	N	normal

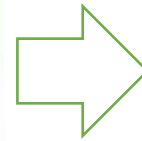


Accuracy = 100%

Attribute	Accuracy
Free	80%
Won	80%
Cash	70%
Free, Won	100%

Forward Selection

ID	Free	Cash	Type
1	Y	Y	spam
2	N	Y	spam
3	N	N	normal
4	N	N	normal
5	Y	N	spam
6	Y	N	spam
7	N	N	normal
8	N	N	spam
9	N	N	normal
10	N	N	normal

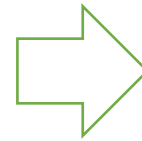


Accuracy = 90%

Attribute	Accuracy
Free	80%
Won	80%
Cash	70%
Free, Won	100%
Free, Cash	90%

Forward Selection

ID	Free	Won	Cash	Type
1	Y	Y	Y	spam
2	N	Y	Y	spam
3	N	N	N	normal
4	N	N	N	normal
5	Y	N	N	spam
6	Y	N	N	spam
7	N	N	N	normal
8	N	Y	N	spam
9	N	N	N	normal
10	N	N	N	normal



Accuracy = 100%

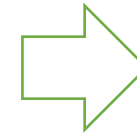
Attribute	Accuracy
Free	80%
Won	80%
Cash	70%
Free, Won	100%
Free, Cash	90%
Free, Won, Cash	100%

Backward Elimination

- Starts with the full set of attributes
- At each step, removes the worst attribute remaining in the set

Backward Elimination

ID	Free	Won	Cash	Type
1	Y	Y	Y	spam
2	N	Y	Y	spam
3	N	N	N	normal
4	N	N	N	normal
5	Y	N	N	spam
6	Y	N	N	spam
7	N	N	N	normal
8	N	Y	N	spam
9	N	N	N	normal
10	N	N	N	normal

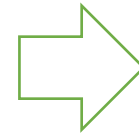


Accuracy = 100%

Attribute	Accuracy
Free, Won, Cash	100%

Backward Elimination

ID	Won	Cash	Type
1	Y	Y	spam
2	Y	Y	spam
3	N	N	normal
4	N	N	normal
5	N	N	spam
6	N	N	spam
7	N	N	normal
8	Y	N	spam
9	N	N	normal
10	N	N	normal

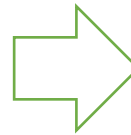


Accuracy = 80%

Attribute	Accuracy
Free, Won, Cash	100%
Won, Cash	80%
Free, Cash	90%
Free, Won	100%

Backward Elimination

ID	Won	Type
1	Y	spam
2	Y	spam
3	N	normal
4	N	normal
5	N	spam
6	N	spam
7	N	normal
8	Y	spam
9	N	normal
10	N	normal

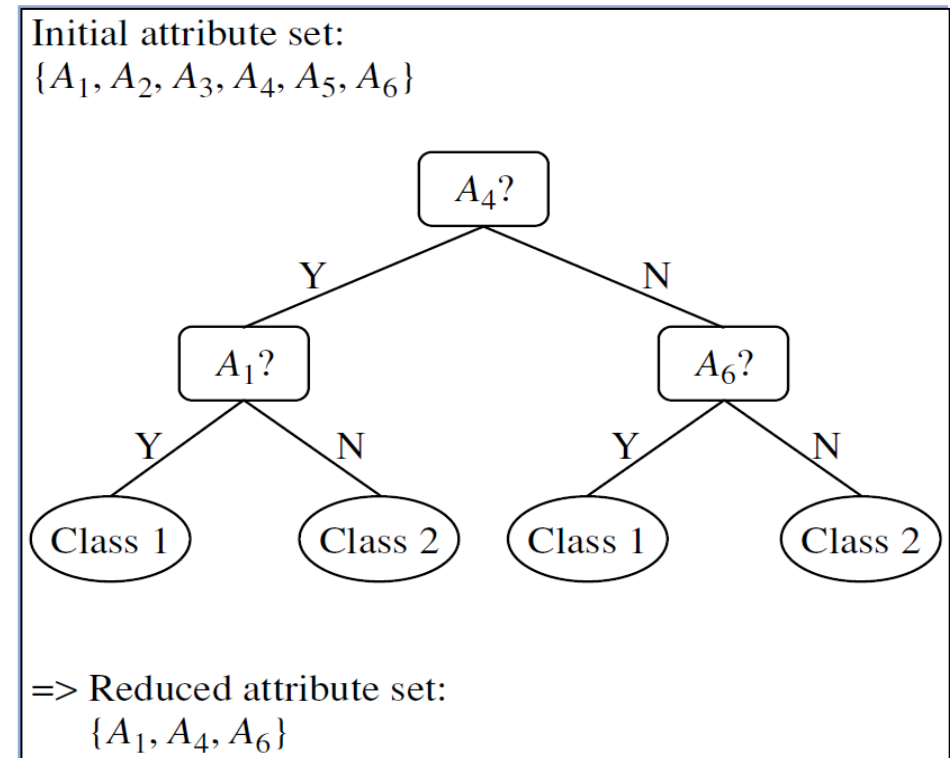


Accuracy = 80%

Attribute	Accuracy
Free, Won, Cash	100%
Won, Cash	80%
Free, Cash	90%
Free, Won	100%
Won	80%
Free	80%

Decision-Tree Induction

- All attributes that do not appear in the tree are assumed to be irrelevant
- The set of attributes appearing in the tree form the reduced subset of attributes

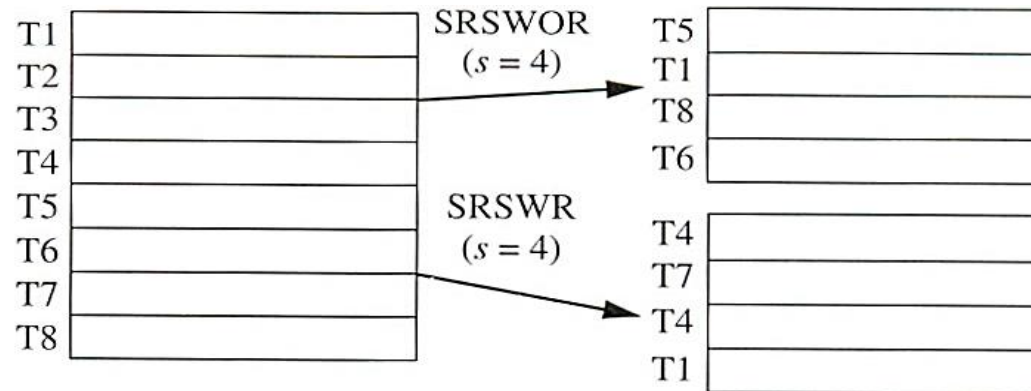


Numerosity Reduction

- Reduce data volume by choosing alternative, smaller forms of data representation
- Parametric methods
 - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - Example: Regression
- Non-parametric methods
 - Do not assume models
 - Major families: histograms, clustering, sampling

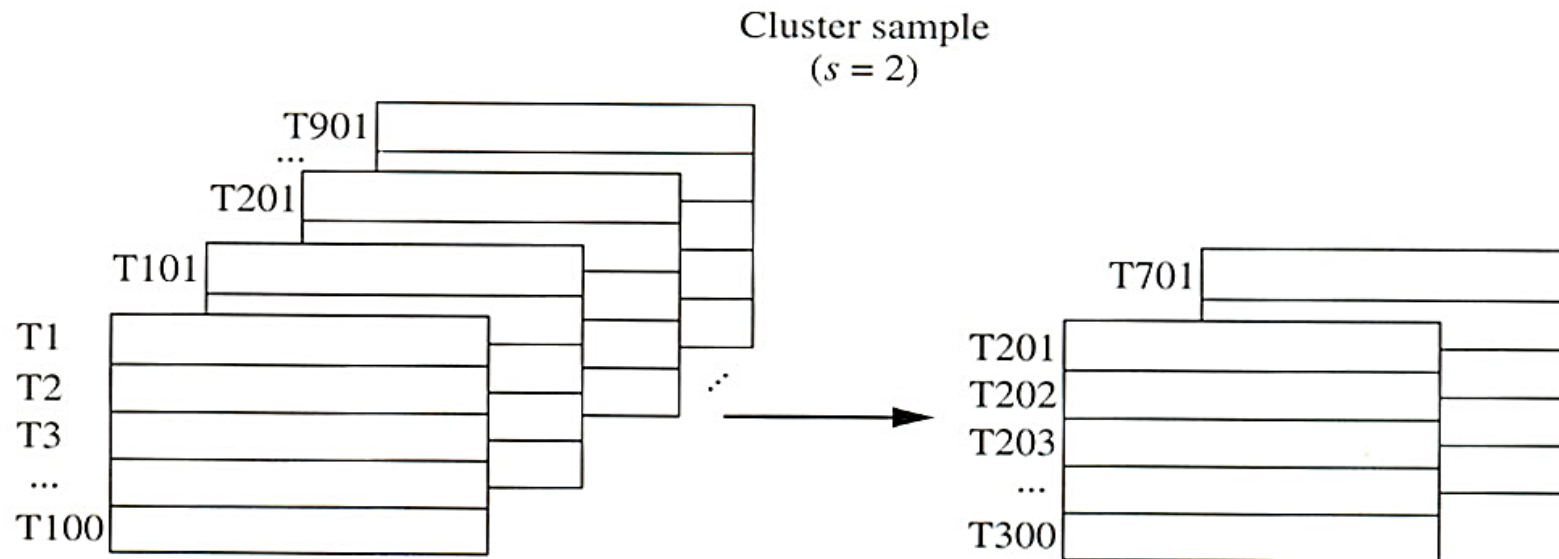
Numerosity Reduction: Sampling

- Sampling
 - obtaining a small sample s to represent the whole data set N
 - Simple Random Sample Without Replacement (SRSWOR)
 - The probability of drawing any tuple in D is $1/N$
 - Simple Random Sample With Replacement (SRSWR)



Numerosity Reduction: Sampling

- Sampling
 - Cluster Sample
 - If the tuples in D are grouped into M mutually disjoint cluster, then an SRS of m clusters can be obtained, where $m < M$



Numerosity Reduction: Sampling

- Sampling
 - Simple random sampling may have very poor performance in the presence of skew
 - Stratified Sample
 - If D is divided into mutually disjoint part called **strata**, a stratified sample of D is generated by obtaining an SRS at each **stratum**

Stratified sample
(according to *age*)

T38	youth
T256	youth
T307	youth
T391	youth
T96	middle_aged
T117	middle_aged
T138	middle_aged
T263	middle_aged
T290	middle_aged
T308	middle_aged
T326	middle_aged
T387	middle_aged
T69	senior
T284	senior

T38	youth
T391	youth
T117	middle_aged
T138	middle_aged
T290	middle_aged
T326	middle_aged
T69	senior

Data Transformation

- Normalization: scaled to fall within a small, specified range
- Attribute/feature construction
 - New attributes constructed from the given ones
 - Add attribute: Area from $W \times H$
- Encoding categorical feature (creating dummies)
- Discretization

Data Transformation: Normalization

- Why normalization?
 - To suit for a certain mining algorithm e.g., Neural network classifier do better with numeric data scaled to $[0..1]$
 - For distance-based algorithms an attribute with a wide range of values are likely to outweigh attribute with smaller ranges
- Normalization methods
 - Min-Max Normalization
 - Decimal Scaling
 - Z-Score
 - Sigmoidal

Normalization: Min-Max Normalization

- Min-max normalization: to $[\text{new_min}_A, \text{new_max}_A]$

$$v' = \frac{v - \text{min}_A}{\text{max}_A - \text{min}_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- Limitation of Min-max normalization
 - min and max values are unknown
 - Not suitable for outlier dataset
- Ex. Let income range \$12,000 to \$98,000 normalized to $[0, 1]$.
 - Then \$73,600 is mapped to

$$v' = \frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

Normalization: Min-Max Normalization

	A	B
1	cus_id	income
2	1	31,330
3	2	97,700
4	3	47,500
5	4	81,000
6	5	36,480
7	6	61,400
8	7	70,800
9	8	88,400
10	9	69,000
11	10	78,000

Min-max normalization
[0,1]



	A	B
1	cus_id	income
2	1	0.000
3	2	1.000
4	3	0.244
5	4	0.748
6	5	0.078
7	6	0.453
8	7	0.595
9	8	0.860
10	9	0.568
11	10	0.703

Normalization: Decimal Scaling

- Normalized by moving the decimal point to values of attribute A
- The number of decimal points moved depends on the maximum absolute value of A

$$v' = \frac{v}{10^j}$$

Where j is the smallest integer such that $\text{Max}(|v'|) < 1$

- Ex. Let attribute A range -999 to 997 normalized to decimal value of -999

$$|-999| = 999, j = 3$$

$$v' = \frac{-999}{10^3} = -0.999$$

Normalization: Decimal Scaling

	A	B
1	cus_id	income
2	1	31,330
3	2	97,700
4	3	47,500
5	4	81,000
6	5	36,480
7	6	61,400
8	7	70,800
9	8	88,400
10	9	69,000
11	10	78,000

Decimal Scaling



$$v' = \frac{v}{10^5}$$

	A	B
1	cus_id	income
2	1	0.3133
3	2	0.9770
4	3	0.4750
5	4	0.8100
6	5	0.3648
7	6	0.6140
8	7	0.7080
9	8	0.8840
10	9	0.6900
11	10	0.7800

Normalization: Z-Score

- Normalized based on the mean and standard deviation of attribute A
- Zero-mean normalization and normalize *std* to 1
- Z-Score is useful when
 - Actual min and max of attributes are unknown
 - There are outliers that dominate the min-max normalization

$$v' = \frac{v - \text{mean}_A}{\text{stand_dev}_A} \quad \text{mean}_A = \frac{\sum A}{n} \quad \text{stand_dev}_A = \sqrt{\frac{\sum (A - \text{mean}_A)^2}{n - 1}}$$

- Ex. Let income range \$12,000 to \$98,000 normalized by Z-score
 - Let $\mu = 54,000$, $\sigma = 16,000$
 - Then \$73,600 is mapped to

$$v' = \frac{73,600 - 54,000}{16,000} = 1.225$$

Normalization: Z-Score

	A	B
1	cus_id	income
2	1	31,330
3	2	97,700
4	3	47,500
5	4	81,000
6	5	36,480
7	6	61,400
8	7	70,800
9	8	88,400
10	9	69,000
11	10	78,000

Z-Score normalization



Mean=66,161

	A	B
1	cus_id	income
2	1	-1.585
3	2	1.436
4	3	-0.849
5	4	0.675
6	5	-1.351
7	6	-0.217
8	7	0.211
9	8	1.012
10	9	0.129
11	10	0.539

Normalization: Sigmoidal

- Normalize input into $[-1, 1]$
- Keeping the outliers

$$v' = \frac{1 - e^{-\alpha}}{1 + e^{-\alpha}} \quad \text{where} \quad \alpha = \frac{y - \text{mean}_A}{\text{stand_dev}_A}$$

Encoding Categorical Feature

- A categorical attribute can be turned into dummy variables
- Transforms each categorical attribute with n categories possible values into n binary attributes, with one of them 1, and all others 0 (called **one hot encoding**)

Label Encoding

Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50



One Hot Encoding

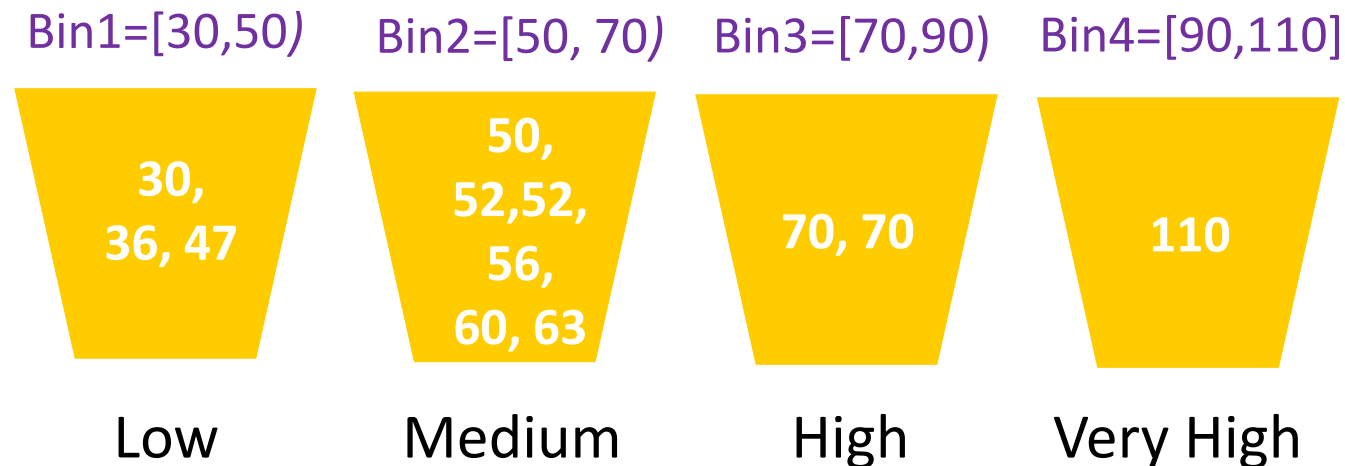
Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50

Discretization

- Discretization:
 - Divide the range of a continuous attribute into intervals
 - Some classification algorithms only accept categorical attributes.
 - Prepare for further analysis
- Typical methods: All the methods can be applied recursively
 - Binning
 - Histogram analysis
 - Clustering analysis
 - Entropy-based discretization

Discretization by Binning

- Example of discretization by binning
- Sorted data for income(in thousands of dollars): 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110
- Partition them into 4 bins by **equal-width**
 - $W = (110 - 30) / 4 = 20$
- Partition into bins:



Discretization by Binning

- Discretization of income

	A	B
1	cus_id	income
2	1	30000
3	2	36000
4	3	47000
5	4	50000
6	5	52000
7	6	52000
8	7	56000
9	8	60000
10	9	63000
11	10	70000
12	11	70000
13	12	110000

discretization



	A	B
1	cus_id	income_level
2	1	Low
3	2	Low
4	3	Low
5	4	Medium
6	5	Medium
7	6	Medium
8	7	Medium
9	8	Medium
10	9	Medium
11	10	High
12	11	High
13	12	Very High